WORKED SOLUTION

STEP II 2002, Question 1

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q1

The key idea

The double-angle formula turns $1-\cos 2\theta$ into $2\sin^2\theta$, and $\operatorname{cosec}^2\theta$ is something we can integrate on sight. The suggested substitution $x=\sin 2\theta$ is designed so that $\sqrt{1-x^2}$ becomes $\cos 2\theta$, which pulls the second integral back to the first. For the last integral, the limits 1 and $2/\sqrt{3}$ are the reciprocals of the limits 1 and $\sqrt{3}/2$, which strongly suggests trying $y=1/x$.

Solution

The first integral

Since $\cos 2\theta = 1-2\sin^2\theta$, we have $1-\cos 2\theta = 2\sin^2\theta$, which is positive on $[\frac{1}{6}\pi, \frac{1}{4}\pi]$, so the integrand is well behaved there. Using $\frac{\mathrm{d}}{\mathrm{d}\theta}(\cot\theta) = -\operatorname{cosec}^2\theta$,

$$\begin{aligned}\int_{\pi/6}^{\pi/4} \frac{1}{1-\cos 2\theta}\,\mathrm{d}\theta &= \frac{1}{2}\int_{\pi/6}^{\pi/4} \operatorname{cosec}^2\theta\,\mathrm{d}\theta = \frac{1}{2}\Big[-\cot\theta\Big]_{\pi/6}^{\pi/4} \\ &= \frac{1}{2}\left(\cot\tfrac{\pi}{6} - \cot\tfrac{\pi}{4}\right) = \frac{1}{2}\left(\sqrt{3}-1\right) = \frac{\sqrt{3}}{2} - \frac{1}{2},\end{aligned}$$

as required.

The second integral

Put $x=\sin 2\theta$, so $\mathrm{d}x = 2\cos 2\theta\,\mathrm{d}\theta$. When $\theta = \frac{1}{6}\pi$, $x = \sin\frac{1}{3}\pi = \frac{\sqrt{3}}{2}$, and when $\theta = \frac{1}{4}\pi$, $x = \sin\frac{1}{2}\pi = 1$; as θ runs over $[\frac{1}{6}\pi, \frac{1}{4}\pi]$, x increases steadily from $\frac{\sqrt{3}}{2}$ to 1.

? **Ask yourself** Is $\sqrt{1-x^2}$ equal to $\cos 2\theta$, or to $-\cos 2\theta$?

We have $\sqrt{1-x^2} = \sqrt{\cos^2 2\theta} = |\cos 2\theta|$. On our range $2\theta \in [\frac{1}{3}\pi, \frac{1}{2}\pi]$, where $\cos 2\theta \geq 0$,

so $\sqrt{1-x^2} = \cos 2\theta$. Therefore

$$\int_{\sqrt{3}/2}^{1} \frac{1}{1-\sqrt{1-x^2}}\,\mathrm{d}x = \int_{\pi/6}^{\pi/4} \frac{2\cos 2\theta}{1-\cos 2\theta}\,\mathrm{d}\theta.$$

To bring in the first integral, write the numerator in terms of the denominator:

$$\frac{\cos 2\theta}{1-\cos 2\theta} = \frac{1-(1-\cos 2\theta)}{1-\cos 2\theta} = \frac{1}{1-\cos 2\theta} - 1.$$

Hence, using the first result,

$$\begin{aligned}\int_{\sqrt{3}/2}^{1} \frac{1}{1-\sqrt{1-x^2}}\,\mathrm{d}x &= 2\int_{\pi/6}^{\pi/4} \frac{1}{1-\cos 2\theta}\,\mathrm{d}\theta - 2\int_{\pi/6}^{\pi/4} 1\,\mathrm{d}\theta \\ &= 2\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right) - 2\left(\frac{\pi}{4} - \frac{\pi}{6}\right) \\ &= \sqrt{3} - 1 - \frac{\pi}{6},\end{aligned}$$

as required.

The final integral

Put $y = 1/x$, so $\mathrm{d}y = -\mathrm{d}x/x^2$. When $y = 1$, $x = 1$, and when $y = 2/\sqrt{3}$, $x = \sqrt{3}/2$. On this range $x > 0$, so

$$\sqrt{y^2-1} = \sqrt{\frac{1-x^2}{x^2}} = \frac{\sqrt{1-x^2}}{x},$$
$$y\left(y - \sqrt{y^2-1}\right) = \frac{1}{x}\cdot\frac{1-\sqrt{1-x^2}}{x} = \frac{1-\sqrt{1-x^2}}{x^2}.$$

Ask yourself Where does the minus sign from $\mathrm{d}y = -\mathrm{d}x/x^2$ go?

The substitution reverses the order of the limits, and swapping them back absorbs the minus sign:

$$\begin{aligned}\int_{1}^{2/\sqrt{3}} \frac{1}{y\left(y-\sqrt{y^2-1}\right)}\,\mathrm{d}y &= \int_{1}^{\sqrt{3}/2} \frac{x^2}{1-\sqrt{1-x^2}}\cdot\left(-\frac{1}{x^2}\right)\mathrm{d}x \\ &= \int_{\sqrt{3}/2}^{1} \frac{1}{1-\sqrt{1-x^2}}\,\mathrm{d}x \\ &= \sqrt{3} - 1 - \frac{\pi}{6}.\end{aligned}$$

The factors of x^2 cancel exactly, which is why the extra factor of y was put in the denominator.

Checks

- The integrands of the second and third integrals are both positive, so the answer must be positive. Indeed $\sqrt{3} - 1 - \frac{\pi}{6} \approx 0.73205 - 0.52360 = 0.20845$.
- Simpson's rule on each of the three integrals gives $0.366025\ldots$ for the first (matching $\frac{\sqrt{3}}{2} - \frac{1}{2}$) and $0.208452\ldots$ for the second and third (matching $\sqrt{3} - 1 - \frac{\pi}{6}$).
- A rough estimate: in the second integral the interval has length $1 - \frac{\sqrt{3}}{2} \approx 0.134$, and the integrand runs from 2 (at $x = \frac{\sqrt{3}}{2}$) down to 1 (at $x = 1$), so the integral should lie between 0.134 and 0.268. It does.

Common slips

- Writing $\sqrt{1 - \sin^2 2\theta} = \cos 2\theta$ without checking the sign of $\cos 2\theta$ on the range.
- Forgetting to change the limits, or losing the minus sign when the limits reverse under $y = 1/x$.
- Evaluating $\left[-\cot\theta \right]$ as $\cot\frac{\pi}{4} - \cot\frac{\pi}{6}$, which gives the right size but the wrong sign.

Beyond STEP

The final part is a small example of a general principle: a substitution is a change of variables that can map one integral exactly onto another, and spotting reciprocal limits is a reliable hint that $y = 1/x$ will do it. In a first analysis course, substitution is justified rigorously from the chain rule and the fundamental theorem of calculus, with care about where the substitution is monotonic and the integrand continuous, exactly the conditions we checked here. Abbott's *Understanding Analysis* gives a clear account of the theory behind these manipulations.

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